

Toxic Threat Zone

ALOHA® 5.4.7



Time: July 16, 2025 1339 hours PDT (using computer's clock)

Chemical Name: PROPANE

Wind: 1.55125 meters/second from 271.4° true at 10 meters

THREAT ZONE:

Model Run: Heavy Gas

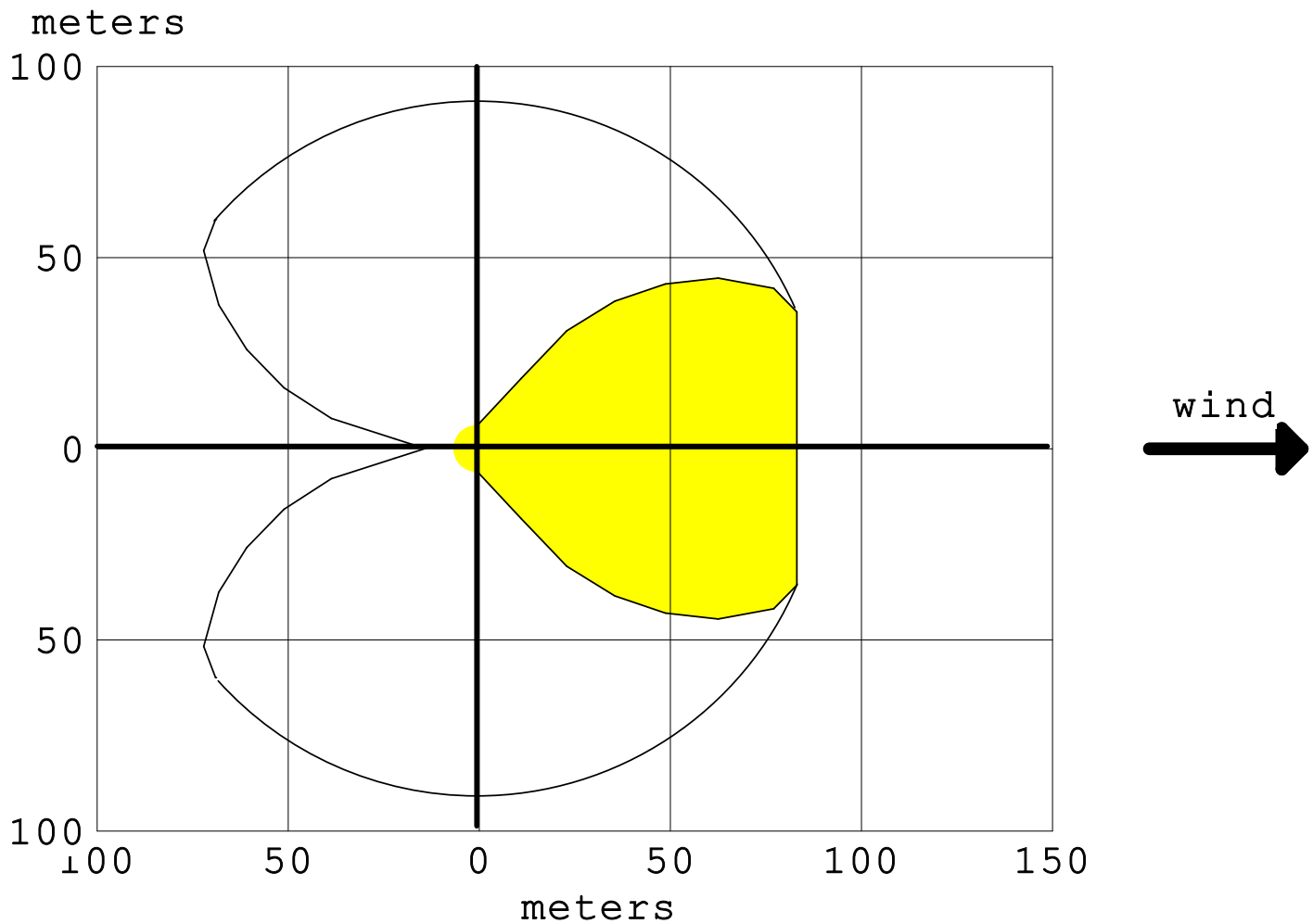
Red : 34 meters --- (33000 ppm = AEGL-3 [60 min])

Note: Threat zone was not drawn because effects of near-field patchiness make dispersion predictions less reliable for short distances.

Orange: 47 meters --- (17000 ppm = AEGL-2 [60 min])

Note: Threat zone was not drawn because effects of near-field patchiness make dispersion predictions less reliable for short distances.

Yellow: 84 meters --- (5500 ppm = AEGL-1 [60 min])



greater than 33000 ppm (AEGL-3 [60 min]) (not drawn)



greater than 17000 ppm (AEGL-2 [60 min]) (not drawn)



greater than 5500 ppm (AEGL-1 [60 min])



wind direction confidence lines

Text Summary

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SITE DATA:

Location: TULARE, CALIFORNIA, USA, CALIFORNIA
Building Air Exchanges Per Hour: 0.28 (unsheltered single storied)
Time: July 16, 2025 1339 hours PDT (using computer's clock)

CHEMICAL DATA:

Chemical Name: PROPANE
CAS Number: 74-98-6 Molecular Weight: 44.10 g/mol
AEGL-1 (60 min): 5500 ppm AEGL-2 (60 min): 17000 ppm AEGL-3 (60 min):
33000 ppm
IDLH: 2100 ppm LEL: 21000 ppm UEL: 95000 ppm
Ambient Boiling Point: -42.3° C
Vapor Pressure at Ambient Temperature: greater than 1 atm
Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)

Wind: 1.55125 meters/second from 271.4° true at 10 meters
Ground Roughness: open country Cloud Cover: 4 tenths
Air Temperature: 18.54° C
Stability Class: D (user override)
No Inversion Height Relative Humidity: 54%

SOURCE STRENGTH:

Leak from hole in vertical cylindrical tank
Flammable chemical escaping from tank (not burning)
Tank Diameter: 5.8 meters Tank Length: 12.9 meters
Tank Volume: 341 cubic meters
Tank contains liquid Internal Temperature: 18.54° C
Chemical Mass in Tank: 146,073 kilograms
Tank is 85% full
Circular Opening Diameter: 1 centimeters
Opening is 12.9 meters from tank bottom
Release Duration: ALOHA limited the duration to 1 hour
Max Average Sustained Release Rate: 70.7 kilograms/min
(averaged over a minute or more)
Total Amount Released: 4,213 kilograms
Note: The chemical escaped as a mixture of gas and aerosol (two phase flow).

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